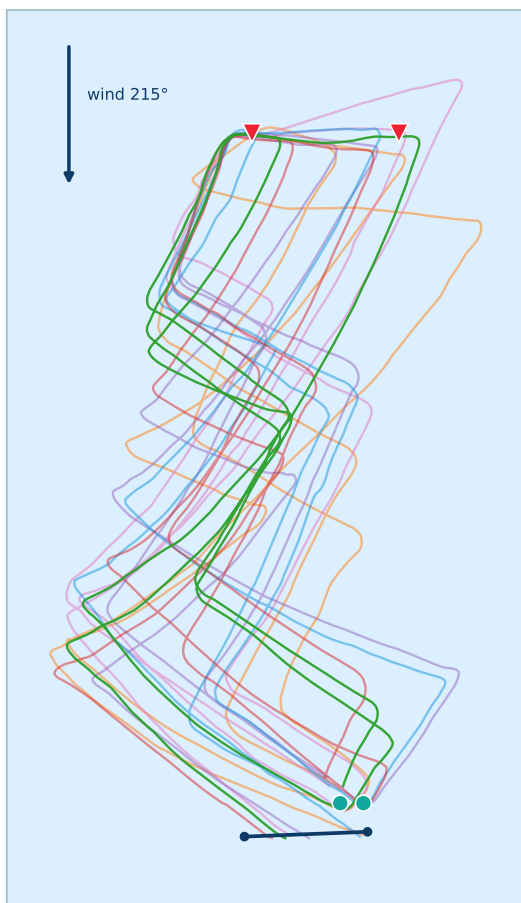


MOTH — Race 5

Lago di Garda, Malcesine · 4 Jul 2026 · Start 15:18 (local)

Race course — all boats



Fleet

- AUS 5250 *
- CRO 5087 *
- GER 5248
- ESP 56
- AUS 5120 *
- USA 4846

Race winner

- AUS 5250 * finished first in this race.
- 2nd: CRO 5087 * 3rd: GER 5248
- Afternoon breeze; W-L-W-L course.
- Wind, marks and course axis estimated from GPS (no wind sensor in logs).

Instrument note (*)

- AUS 5250, CRO 5087, AUS 5120: speed sensor mounted rotated. Heel & heel-stability are rebuilt from the boat's gravity + course — an accurate estimate, not exact to the degree.

Positions at the marks — gains (green) / losses (red)

Finish	Boat	M1	M2	M3	Fin	Δ M1→M2	Δ M2→M3	Δ M3→Fin	Net
1	AUS 5250 *	2	1	1	1	+1	+0	+0	+1
2	CRO 5087 *	3	2	2	2	+1	+0	+0	+1
3	GER 5248	4	3	3	3	+1	+0	+0	+1
4	ESP 56	5	4	5	4	+1	-1	+1	+1
5	AUS 5120 *	1	5	4	5	-4	+1	-1	-4
6	USA 4846	6	6	6	6	+0	+0	+0	+0

M1 = windward 1 · M2 = leeward 1 · M3 = windward 2 · Fin = finish. Positive Δ (green) = places gained on that leg.



Fin	Boat	Line pos	Bias loss (m)	DTL gun	DTL -10s	SOG gun	SOG -10s	TWA gun	SOG 1'	Heel stab 1'	VMG 1'
1	AUS 5250 *	2-3	18	10	35	24.2	19.0	73	20.8	8.1	13.5
2	CRO 5087 *	3-1	12	7	21	23.1	19.9	81	20.8	3.0	12.6
3	GER 5248	1-1	50	13	38	19.8	19.0	66	17.0	9.5	9.8
4	ESP 56	2-2	28	12	44	20.7	17.5	69	19.2	4.9	11.0
5	AUS 5120 *	2-3	21	3	31	22.5	21.1	75	19.7	10.2	12.8
6	USA 4846	1-1	51	8	1	5.2	11.3	63	16.2	10.8	6.1

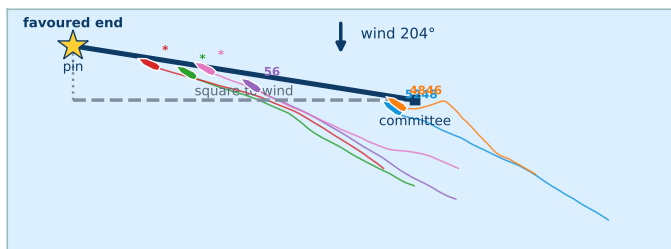
How to read the start

- Line pos: the line is split into 3 thirds, each third into 3 — read third-subpart from the committee end (1-1) to the pin end (3-3).
- Bias loss (m): metres given away at the gun from line position alone vs the favoured (pin) end, from the TWD bias.
- DTL = distance to line (m, negative = over early). 1' = first minute after the gun; heel stability = avg heel change per 10s (lower = steadier).

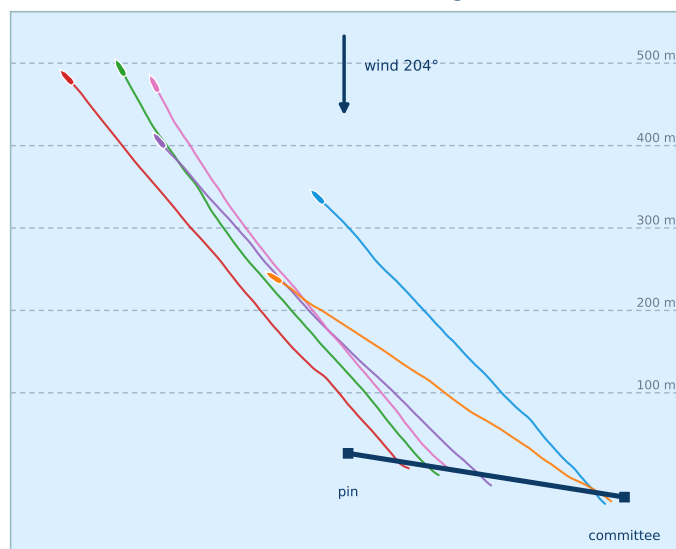
Line bias at the start (vs TWD)

- Start line 340 m, bearing 303°.
- Wind at start ≈ 204°, estimated from the first beat.
- Line bias vs TWD: 9.0° — Pin end favoured.
- Advantage ≈ 53 m (≈ 10.7 boat-lengths) for a boat starting at the pin end.

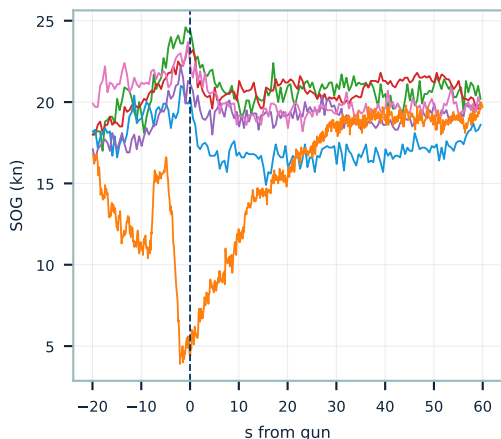
Position at gun



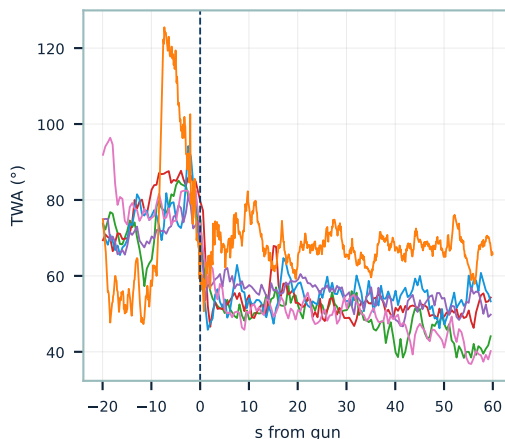
Position 1 min after gun



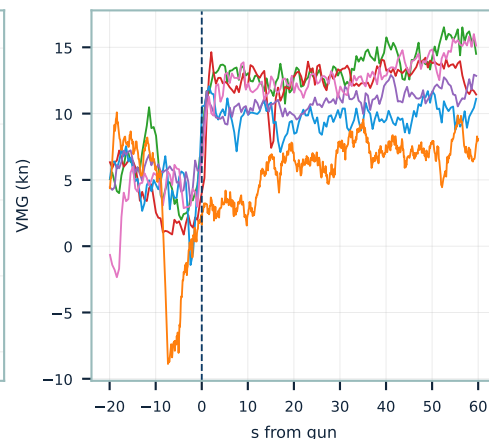
SOG around gun



TWA around gun



VMG around gun

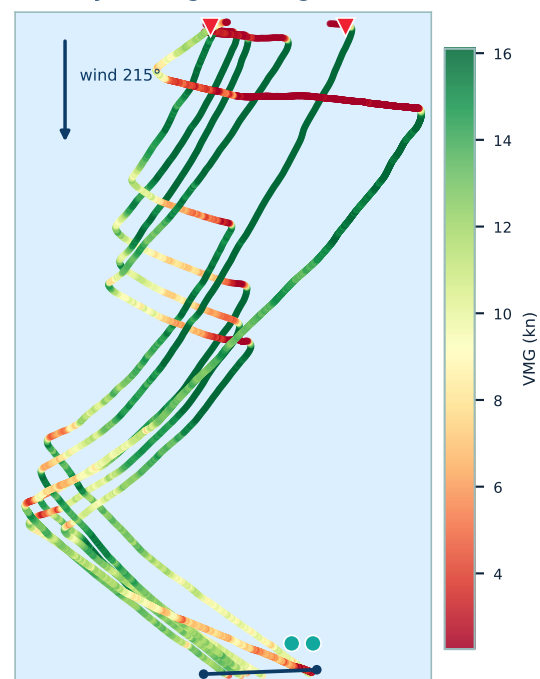




Tracks by boat



Tracks by VMG (green=high, red=low)



— AUS 5250*
 — CRO 5087*
 — GER 5248
 — ESP 56
 — AUS 5120*
 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
5	AUS 5120*	14.39	41	19.25	21.30	23.9	9.1	-1.8	-0.4	2598	262	2
1	AUS 5250*	12.81	47	19.43	21.92	24.1	13.0	+0.1	+0.3	2937	294	4
2	CRO 5087*	12.13	49	18.77	21.60	22.0	8.4	-0.9	+0.6	2979	309	4
3	GER 5248	11.90	45	17.32	19.70	25.7	11.8	+2.8	-0.2	2826	318	4
4	ESP 56	11.09	53	18.25	21.40	23.1	9.6	+4.4	+0.2	3178	339	4
6	USA 4846	8.83	59	17.23	20.72	19.5	7.6	-0.9	-0.2	3802	429	3

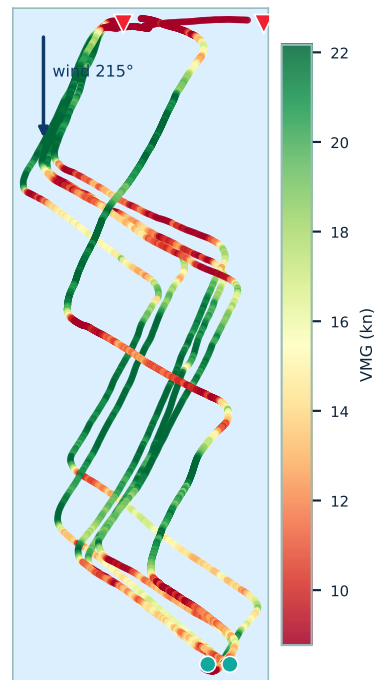
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



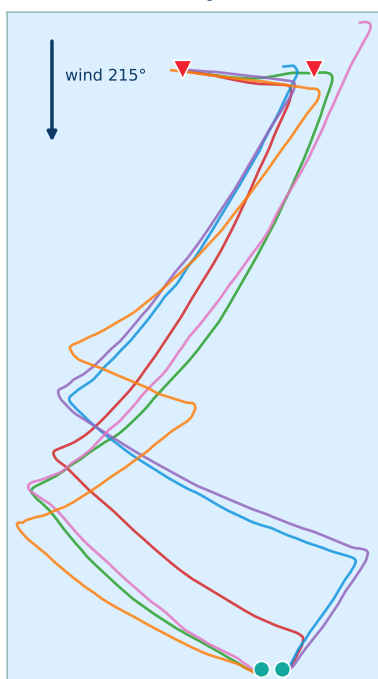
— AUS 5250*
 — CRO 5087*
 — GER 5248
 — ESP 56
 — AUS 5120*
 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
2	CRO 5087*	17.42	141	23.42	25.98	14.4	9.2	+1.9	-0.1	2462	204	3
1	AUS 5250*	16.82	138	23.46	26.00	15.6	9.2	-1.1	+0.8	2579	214	4
3	GER 5248	16.40	140	22.38	24.92	14.4	8.8	-0.2	-0.5	2533	220	3
4	ESP 56	16.28	139	22.64	24.92	9.9	8.2	-1.1	-0.5	2549	218	3
6	USA 4846	15.15	140	20.84	25.30	10.5	7.2	+2.9	+0.3	2590	241	3
5	AUS 5120*	12.48	128	20.09	25.90	13.0	10.7	+4.4	-0.3	3028	298	5

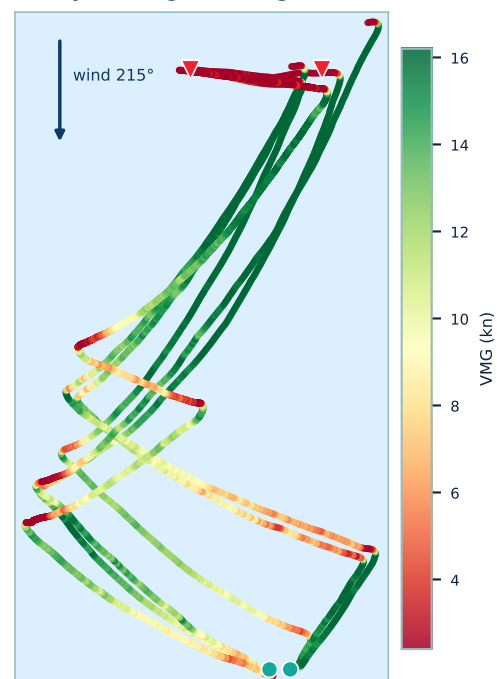
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— AUS 5250*
 — CRO 5087*
 — GER 5248
 — ESP 56
 — AUS 5120*
 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
5	AUS 5120*	13.10	44	18.15	21.41	25.1	11.3	-2.7	+0.2	2782	298	2
1	AUS 5250*	12.27	49	19.84	22.60	25.6	10.4	+1.7	+0.2	2997	293	2
3	GER 5248	12.23	46	18.10	20.10	27.6	13.7	+3.6	-0.3	2743	295	3
2	CRO 5087*	12.23	47	18.64	20.10	22.6	7.4	-0.7	+0.3	2799	292	3
4	ESP 56	11.19	52	18.85	21.04	23.0	11.1	+5.4	-0.3	3095	320	3
6	USA 4846	8.71	58	16.87	20.90	19.1	13.3	-2.6	-0.2	3611	416	4

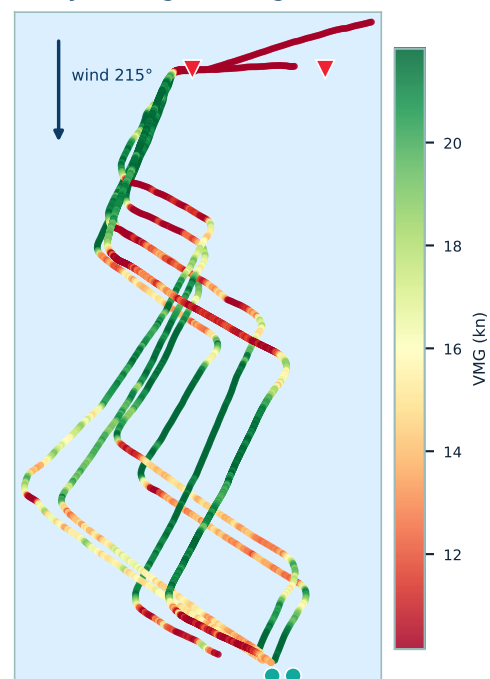
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— AUS 5250*
 — CRO 5087*
 — GER 5248
 — ESP 56
 — AUS 5120*
 — USA 4846

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
1	AUS 5250*	17.52	140	23.94	26.30	16.7	10.2	+0.8	+0.2	2448	199	4
2	CRO 5087*	17.40	141	23.42	26.00	14.6	10.3	+2.3	+0.5	2380	197	4
4	ESP 56	16.88	140	23.25	26.80	12.3	8.9	-1.1	-0.4	2437	203	3
6	USA 4846	16.50	141	22.28	24.90	11.8	5.7	+2.8	+0.2	2437	212	2
3	GER 5248	14.81	135	22.13	26.26	16.4	10.6	+2.4	-0.4	2673	235	3
5	AUS 5120*	14.11	130	23.21	28.30	14.5	9.0	+2.9	-0.6	3197	268	3

Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Finish	Boat	# Tacks	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
1	AUS 5250 *	6	18.6	13.9	11.8	6.8	+20	103	21	23	25.5	40.4
3	GER 5248	7	17.9	13.1	11.0	6.9	+18	116	22	24	26.4	39.1
2	CRO 5087 *	7	19.1	13.7	11.5	7.6	+18	98	19	22	30.0	39.3
4	ESP 56	7	19.9	14.2	11.4	8.6	+18	140	20	22	31.3	41.0
5	AUS 5120 *	4	19.3	13.3	10.7	8.6	+23	115	19	20	32.5	48.1
6	USA 4846	7	16.7	9.7	7.0	9.7	+26	64	12	20	39.0	48.8

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

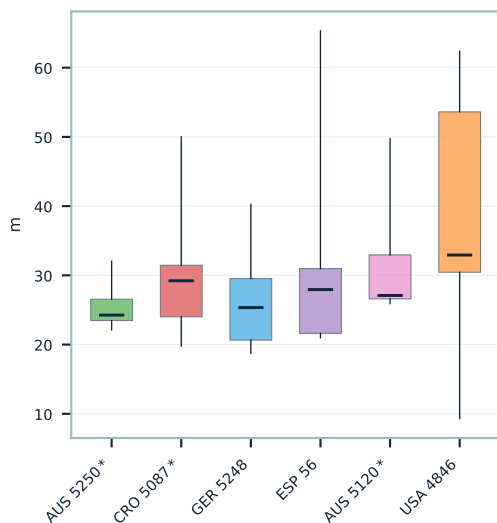
Tacks analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

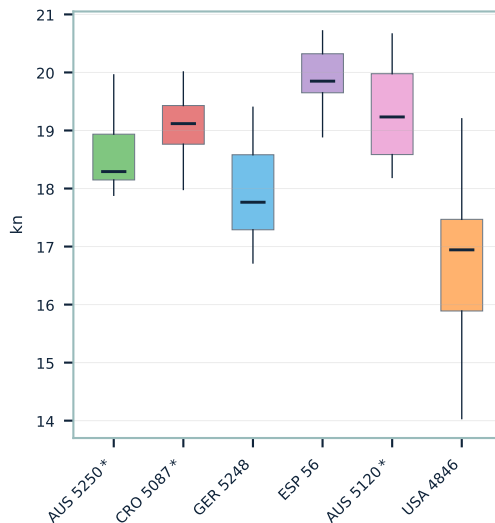
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

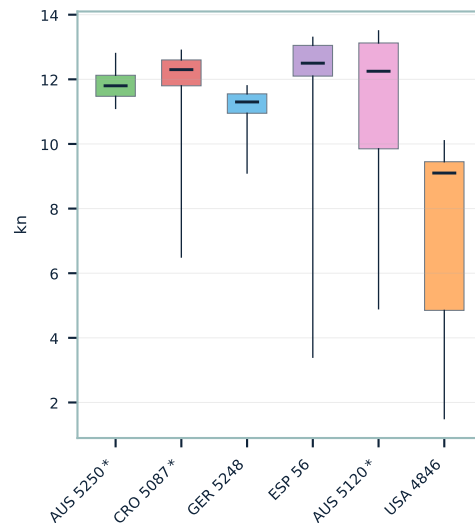
Dist loss (m, lower=better)



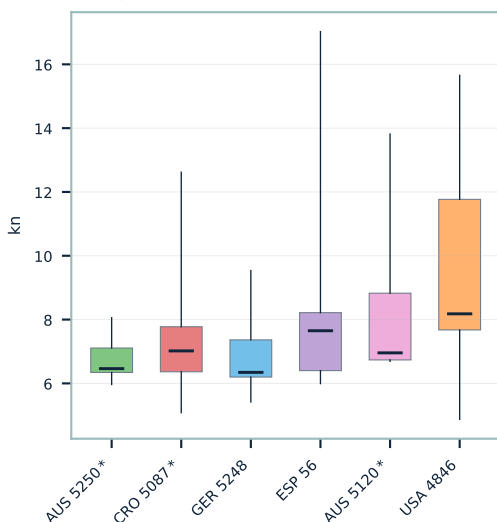
Entry speed (higher = better)



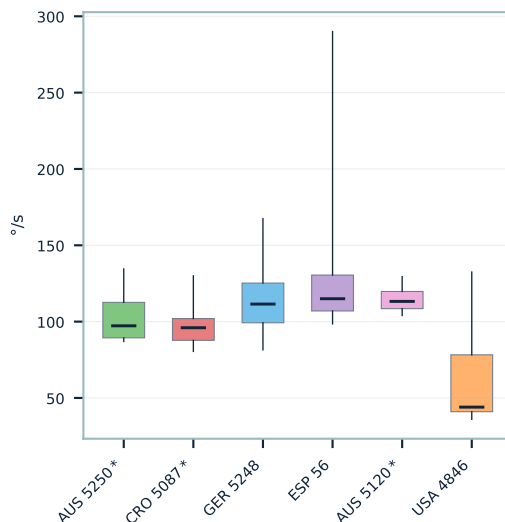
Min speed (higher = better)



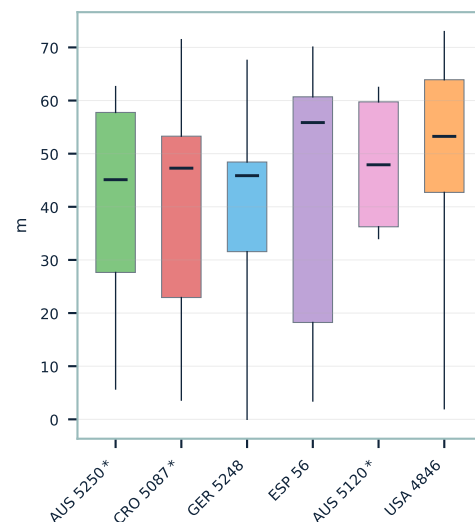
Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)





Finish	Boat	# Gybes	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
1	AUS 5250 *	8	23.8	20.7	18.7	5.1	-2	90	14	14	16.9	19.5
5	AUS 5120 *	8	19.8	16.7	14.7	5.1	-2	164	16	10	17.9	20.7
6	USA 4846	5	20.3	17.0	15.1	5.2	+4	32	8	9	18.3	11.0
4	ESP 56	6	22.5	19.1	17.0	5.5	-5	74	11	8	18.5	23.3
2	CRO 5087 *	7	23.4	19.6	17.8	5.6	-5	66	12	11	21.3	23.9
3	GER 5248	6	22.3	18.3	16.4	5.9	-7	79	12	10	22.0	30.6

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

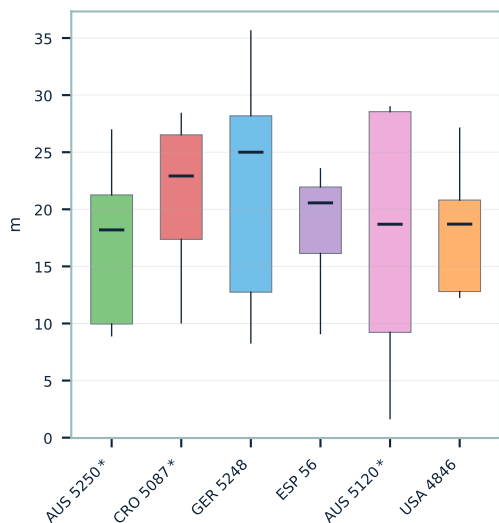
Gybes analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

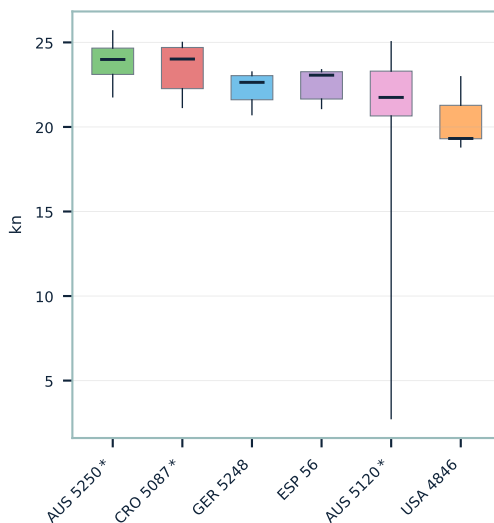
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

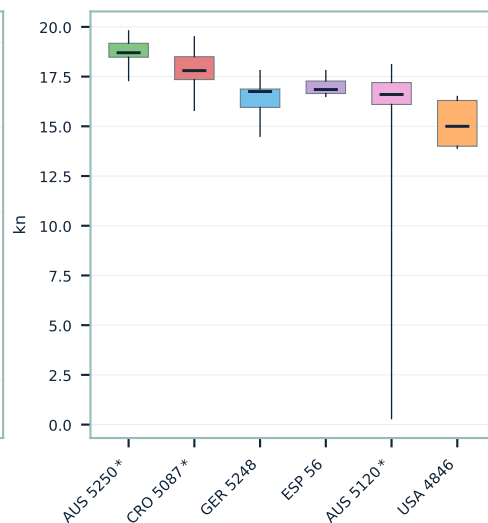
Dist loss (m, lower=better)



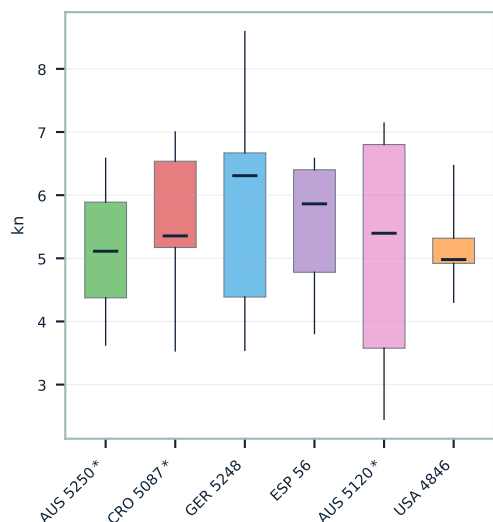
Entry speed (higher = better)



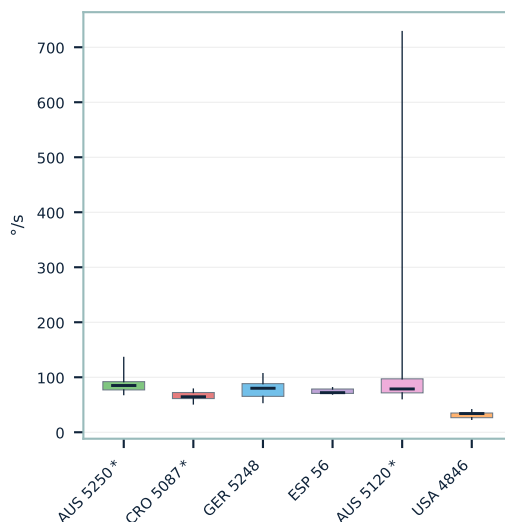
Min speed (higher = better)



Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)

